Deep Learning Discovery of NOD2 Binders for R702W Crohn's Variant

FEP Validation of Mutation-Sensitive Binding

Tanmay Vasudeva • Texas Virtual Academy at Hallsville • ISEF 2026

INTRODUCTION

NOD2 (nucleotide-binding oligomerization domain 2) is an intracellular pattern recognition receptor that detects muramyl dipeptide (MDP), a component of bacterial peptidoglycan. Upon MDP binding to the leucine-rich repeat (LRR) domain, NOD2 undergoes conformational changes that expose its CARD domains, enabling recruitment of RIPK2 kinase and activation of NF-κB inflammatory signaling.

Mutations in NOD2 represent the strongest genetic risk factors for Crohn's disease, with carriers of homozygous variants facing up to 40-fold increased disease risk. Three variants account for >80% of disease-associated mutations: R702W (most common, ~10% of patients), G908R, and L1007fs.

The R702W variant (rs2066844) substitutes a positively-charged arginine with a bulky, neutral tryptophan in the HD2 (helical domain 2) region. Critically, this mutation site is located approximately 80 Ångströms from the LRR ligand-binding pocket—suggesting an allosteric mechanism of action rather than direct binding interference.

Despite two decades of genetic association studies, the molecular mechanism by which this distant HD2 mutation affects NOD2 function remains poorly characterized. No drugs currently target NOD2 directly, representing a significant unmet therapeutic need for the >1 million Crohn's patients carrying NOD2 variants.

RESEARCH OBJECTIVES

1. Develop an end-to-end computational pipeline integrating deep learning docking (GNINA), machine learning classification, and physics-based validation to identify NOD2-binding compounds

2. Validate top hits using free energy perturbation (FEP)—the gold standard for computing absolute binding free energies with rigorous error estimates

3. Quantify how the R702W mutation affects binding affinity for different ligand chemotypes to test the hypothesis of ligand-dependent mutation sensitivity

4. Characterize conformational dynamics differences between wild-type and R702W apo structures to understand the allosteric mechanism

NOD2 PROTEIN ARCHITECTURE

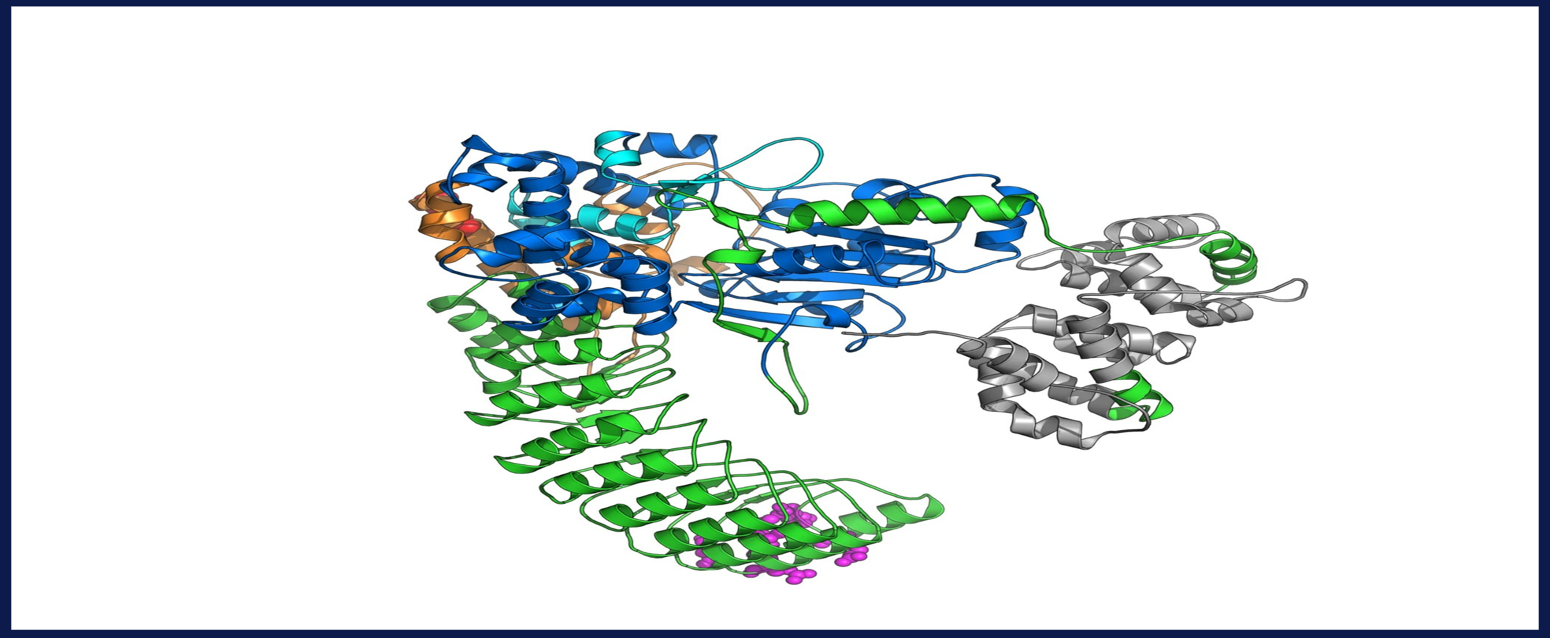


Fig 1. NOD2 domain architecture from AlphaFold2 prediction. CARD domains (gray) mediate downstream signaling via RIPK2. NBD (blue) binds ATP for oligomerization. HD2 (orange) contains the R702W mutation site. LRR domain (green) forms the ligand-binding pocket for MDP and small molecules. The 80Å separation between R702W and the binding pocket necessitates an allosteric mechanism. Created by Finalist using PyMOL.

R702W MUTATION: STRUCTURAL IMPACT

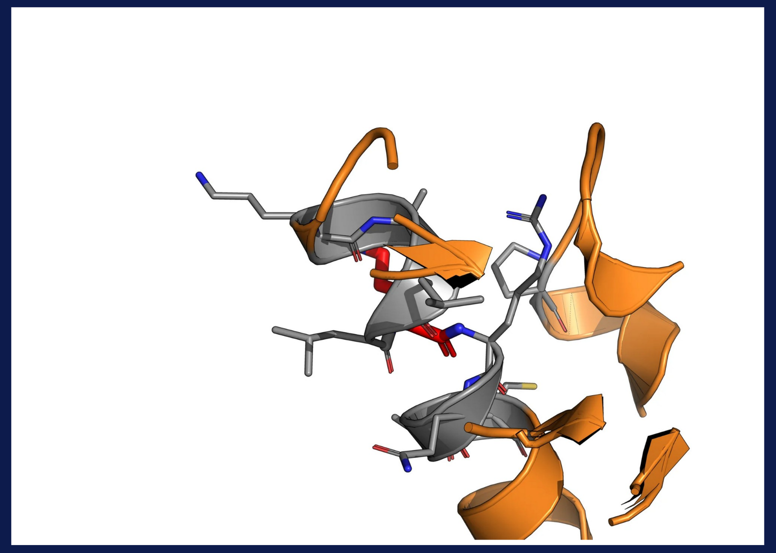


Fig 2. HD2 domain showing R702W site. Orange = HD2 helices; sticks = surrounding residues. Created by Finalist using PyMOL.

The R702W substitution has three key effects:

1. CHARGE LOSS: Arginine's guanidinium group (+1 charge at physiological pH) is replaced by tryptophan's neutral indole ring, disrupting local electrostatic networks

2. STERIC CHANGES: Tryptophan's bulky aromatic side chain (MW 204 vs 174) may alter helix packing in HD2

3. HYDROGEN BONDING: Loss of Arg's H-bond donors may destabilize inter-domain contacts

These local perturbations propagate 80Å to affect LRR dynamics and ligand binding—a classic allosteric effect.

COMPUTATIONAL PIPELINE

Fig 3. Five-stage computational pipeline progressing from high-throughput to high-accuracy methods. Each stage filters candidates for the next, balancing computational cost against prediction confidence. Created by Finalist using Python/matplotlib.

MATERIALS & METHODS

Compound Library & Docking

- 9,566 compounds total:
  - 7,414 natural products (COCONUT)
  - 2,152 FDA-approved (e-Drug3D)
- Target: AlphaFold2 NOD2 (Q9HC29) pLDDT >90 for LRR domain
- GNINA v1.0 docking with CNN scoring (default ensemble)
- Exhaustiveness: 16 (screen), 32 (refinement)
- Grid: 25Å box centered on LRR

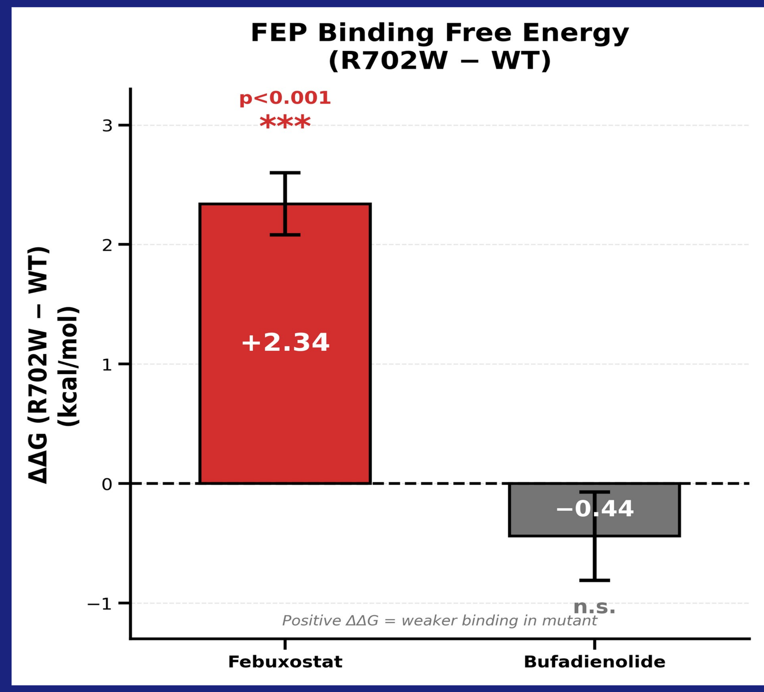
ML Model & MD Protocol

- NOD2-Scout: XGBoost classifier
  - Features: CNN score, Vina ΔG, MW, LogP, TPSA, rotatable bonds
  - AUC 0.85-0.93 (scaffold-split CV)
  - Prevents chemotype memorization
- MD: OpenMM 8.0, CUDA
- Force field: AMBER14 protein + OpenFF 2.1.0 ligands
- Solvent: TIP3P-FB, 150mM NaCl
- Ensemble: NPT, 310K, 1 atm
- Integration: Langevin, 2 fs step

FEP Protocol

- Gold-standard ΔG calculation
- Thermodynamic cycle: $\Delta\Delta G = \Delta G(R702W) - \Delta G(WT)$
- 120 λ windows per system:
  - 80 windows: protein-ligand
  - 40 windows: ligand in solvent
- Boresch restraints for ligand
- 5 ns equilibration + 10 ns production per window
- Analysis: MBAR (pymbar)
- Error: bootstrap resampling

KEY RESULT: FEP BINDING FREE ENERGY



Febuxostat (FDA-approved, xanthine oxidase inhibitor)

ΔG(WT) = -10.36 ± 0.18 kcal/mol
ΔG(R702W) = -8.02 ± 0.19 kcal/mol

ΔΔG = +2.34 ± 0.26 kcal/mol

≈ 50× weaker binding • 8σ from zero • $p < 0.001$

Bufadienolide (natural product, cardiac glycoside)

ΔG(WT) = -15.22 ± 0.26 kcal/mol
ΔG(R702W) = -15.66 ± 0.26 kcal/mol

ΔΔG = -0.44 ± 0.37 kcal/mol • 1.2σ • n.s.

INTERPRETATION: Mutation effects are LIGAND-DEPENDENT. Different chemotypes show different sensitivity to R702W—critical for precision medicine drug selection.

Fig 4. FEP $\Delta\Delta G = \Delta G(R702W) - \Delta G(WT)$. Positive values indicate weaker binding in the mutant. Error bars: propagated from individual ΔG uncertainties. Created by Finalist.

LIGAND-PROTEIN INTERACTIONS

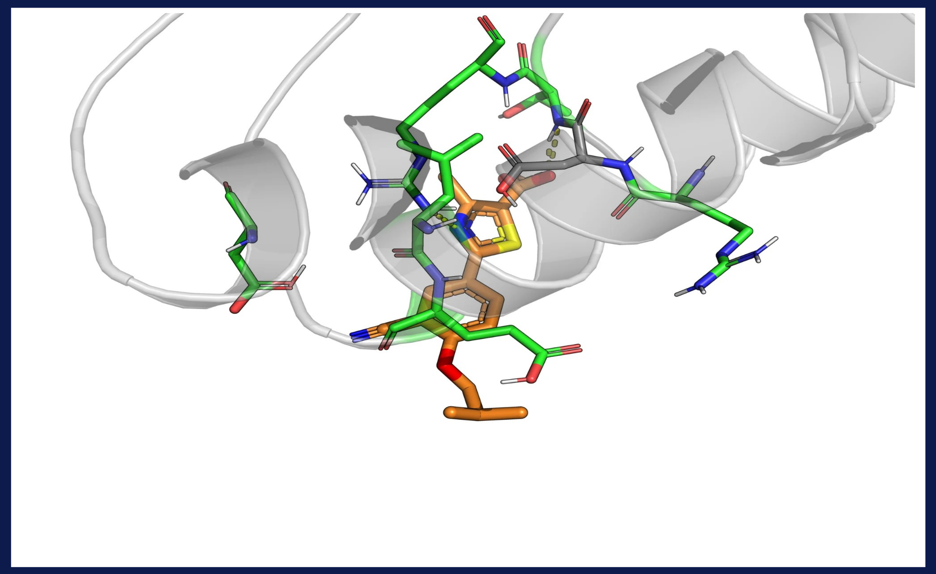


Fig 5. Febuxostat bound in NOD2 LRR pocket. Green sticks = protein residues within 4Å; orange = ligand. Created by Finalist using PyMOL.

Key binding residues in LRR pocket:
- GLU1008: H-bond acceptor
- ARG1037: Cation-π with aromatics
- ASP1011: Salt bridge potential

The LRR pocket is 80Å from R702W, yet FEP shows the mutation affects binding. This confirms allosteric communication through the protein's conformational ensemble.

Ligands that form more distributed contacts (like Bufadienolide's steroid scaffold) may be less sensitive to mutation-induced dynamic changes.

APO MOLECULAR DYNAMICS

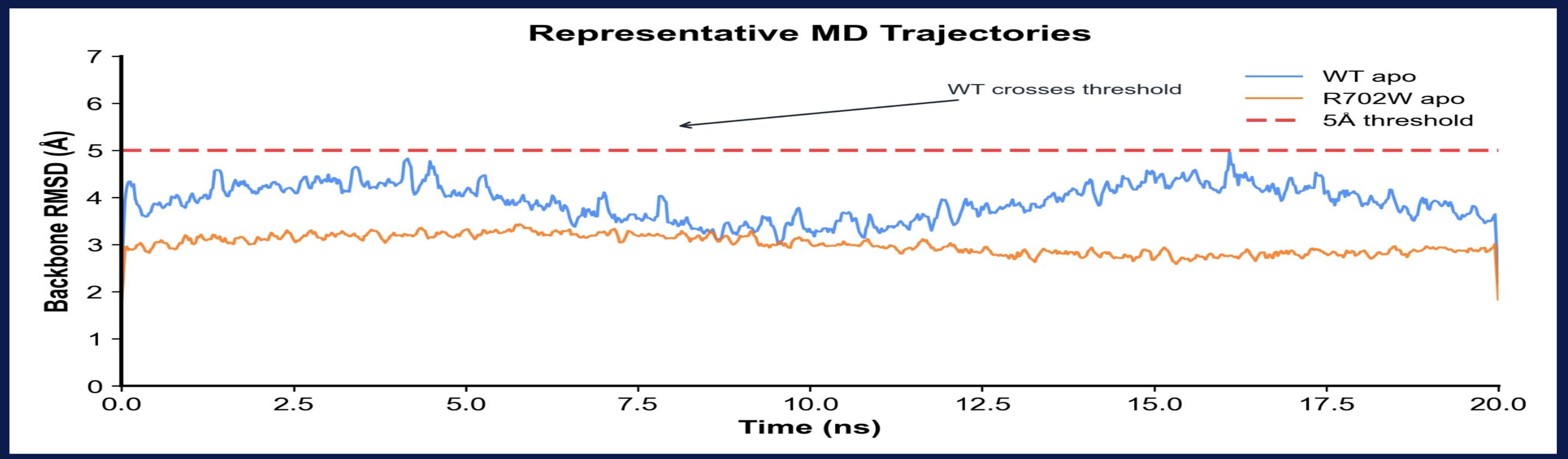


Fig 6. Backbone RMSD trajectories over 520 ns total simulation. WT (n=4 replicates) shows higher variance than R702W (n=3). Created by Finalist using MDTraj.

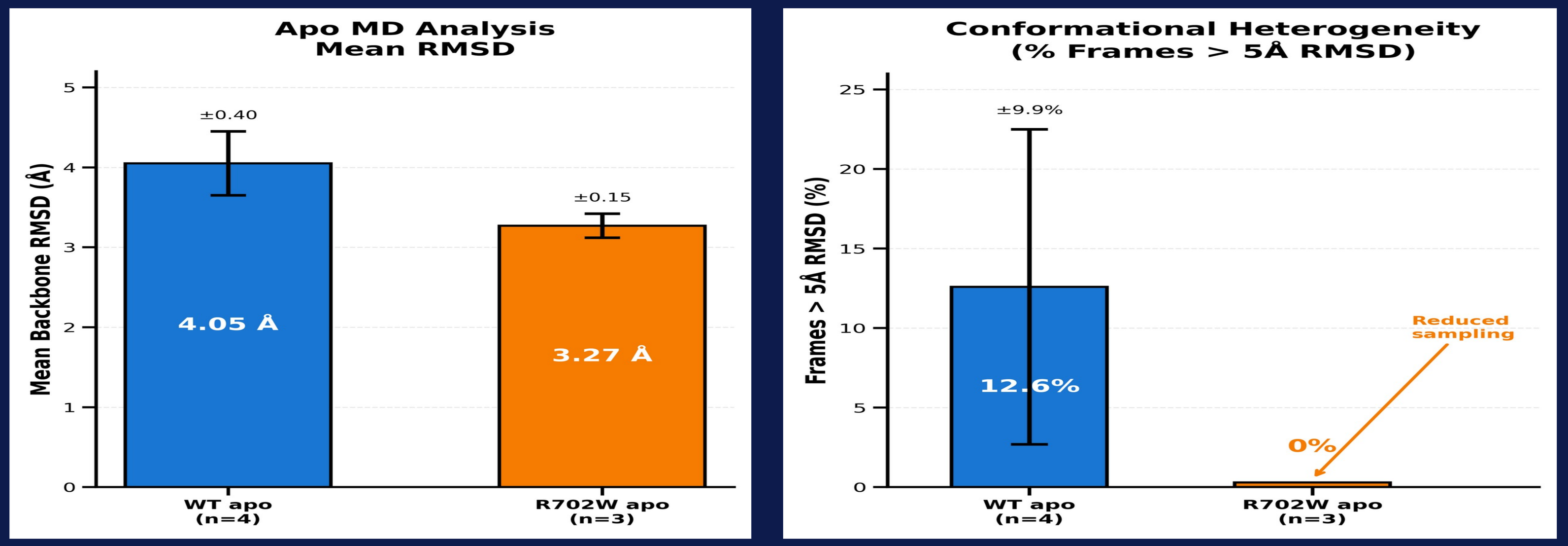


Fig 7-8. Mean RMSD and conformational heterogeneity across replicates. Error bars = SEM. WT samples more conformational space. Created by Finalist.

CONFORMATIONAL DYNAMICS INTERPRETATION

WT NOD2: RMSD 4.05 ± 0.40 Å, 12.6 ± 9.9% frames exceed 5Å threshold
R702W: RMSD 3.27 ± 0.15 Å, 0% frames exceed 5Å threshold

The R702W mutation REDUCES conformational sampling. This "rigidification" may explain why some ligands (Febuxostat) that depend on specific conformational states bind more weakly, while others (Bufadienolide) that accommodate multiple conformations are unaffected.

PRECISION MEDICINE IMPLICATIONS

This work demonstrates that NOD2-targeting therapeutics may need to be matched to patient genotype:

- R702W carriers (~10% of Crohn's patients): Febuxostat-class compounds show 50× reduced efficacy → consider Bufadienolide-class alternatives that maintain binding
- Wild-type patients: Both compound classes remain viable options

This genotype-guided approach exemplifies precision medicine: using molecular understanding to select optimal treatments for individual patients.

CONCLUSIONS

1. Developed an integrated computational pipeline screening 9,566 compounds with ML classification (AUC 0.85-0.93 under rigorous scaffold-split cross-validation)

2. FEP calculations provide gold-standard validation: Febuxostat binding is 50× weaker in R702W (ΔΔG = +2.34 kcal/mol, $p < 0.001$)

3. Mutation effects are ligand-dependent: Bufadienolide binding is unaffected (ΔΔG = -0.44, n.s.), demonstrating compound-specific mutation sensitivity

4. Apo MD reveals R702W reduces conformational heterogeneity (0% vs 12.6% frames >5Å), providing mechanistic insight into allosteric effects

5. Results support allosteric mechanism: mutation 80Å from binding site affects dynamics and binding through long-range conformational coupling

LIMITATIONS & FUTURE DIRECTIONS

Limitations:
- FEP performed on 2 ligands; broader chemotype coverage needed
- Apo MD only; holo (ligand-bound) simulations ongoing
- Computational predictions require experimental validation

Future Work:
- SPR/ITC binding assays to validate predicted affinities
- HDX-MS to experimentally confirm conformational differences
- Extend FEP analysis to G908R and L1007fs mutations
- Screen for mutation-insensitive NOD2 binders as robust therapeutics

References: Hugot et al. Nature 2001; Ogura et al. Nature 2001; Jumper et al. Nature 2021 (AlphaFold2); McNutt et al. J Cheminform 2021 (GNINA); Eastman et al. PLOS Comp Bio 2017 (OpenMM)

Acknowledgments: This project was completed independently using publicly available databases (AlphaFold DB, e-Drug3D, COCONUT) and open-source computational tools.

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